# A Dual-Path Efficient EEG Encoder for Brain-Assisted Target Speaker Extraction

Wang Xiang[1,2], Xue Zhang[3], Bang Zeng[4], Cunhang Fan[3], Juan Liu[1,5], and Ming Li[1,2⋆]

[1] School of Artificial Intelligence, Wuhan University, Wuhan, China
[2] School of Artificial Intelligence, The Chinese University of Hong Kong, Shenzhen, China
[3] School of Computer Science and Technology, Anhui University, Hefei, China
[4] School of Computer Science, Wuhan University, Wuhan, China
[5] School of Cyber Science and Engineering, Wuhan University, Wuhan, China
mingli369@cuhk.edu.cn

**Abstract.** Brain-assisted target speaker extraction can selectively extract the target speaker's speech by decoding the listener's brain activity from electroencephalogram (EEG) signals. However, EEG has low signal-to-noise ratio, high dimensionality, and non-stationarity, making it very challenging to efficiently learn advanced features. To address these limitations, this paper proposes a dual-path Receptance Weighted Key Value-like (RWKV) EEG encoder designed to mitigate the inefficiency and difficulty of modeling long sequences and weak correlations inherent in EEG. Specifically, the encoder overcomes the limitations of traditional EEG encoders by applying cortical interaction and temporal dynamics modules, enabling the model to simultaneously capture multidimensional spatiotemporal information in EEG signals with linear computational complexity. Experiments on three public datasets show that the proposed model significantly outperforms current state-of-the-art methods in multiple evaluation metrics.

**Keywords:** Target speaker extraction · EEG signals · Brain-assisted speech separation · Multi-talker conditions

## 1 Introduction

The human auditory system can selectively attend to a target speaker while suppressing competing sounds in multi-speaker environments, known as the "cocktail party effect"[1]. Target speaker extraction is proposed to emulate this capability by exploiting prior information of the target speaker (e.g., speaker embedding, lip movements, or reference speech), and directly extracts the target speech from a mixture[2,3,4,5,6,7].

Nevertheless, the effectiveness of these cues remains inferior to the selective attention mechanisms of the human auditory system and may be further

⋆ Corresponding author: Ming Li

constrained by practical conditions[8,9,10,11,12,13]. Clues based on speaker embedding or reference speech require additional reference utterance of the target speaker[14,15]. Visual cues based on lip movements and faces require continuous or unobstructed frontal face, which cannot be satisfied in camera-free scenarios for real-world applications[16,17]. Regarding the content clues, there are usability limitations due to privacy and security concerns[18].

The auditory system of the human brain can selectively focus only on sounds of interest from complex acoustic scenes. Recent studies [19,20] have shown that it is possible to locate the target sound source from human brain neural signals, such as electroencephalography signals, and decode which the subject is paying attention to, namely auditory attention decoding or auditory attention neural decoding[21,22]. Therefore, using human brain neural signals as prior information, the target speaker information decoded from EEG signals is applied to target speaker extraction, thereby achieving brain-assisted target speaker extraction.

Although this brain-assisted method has unique advantages, its development is still limited by some key issues. EEG has the characteristics of low signal-to-noise ratio, high-dimensional, non-stationary, and contains semantic information related to auditory attention and a large amount of irrelevant physiological noise signals[23,24,25]. Previous methods using handcrafted features (e.g. time-domain ERPs, frequency-domain power spectra) with classifiers have limited capacity to model complex nonlinear neural dynamics[26]. How to robustly and efficiently extract high-quality and task-related features becomes the focus. Recent approaches[27,13] alleviate feature engineering, but progress is often constrained by a trade-off between representation capacity and computational efficiency: models that better capture long-context dependencies and inter-electrode interactions can become expensive and less stable on noisy EEG, while lightweight designs may underfit the attention-related cues. Therefore, designing an EEG representation learner that is both robust to noise and artifacts and efficient enough to scale with the number of channels and the sequence length remains a key open problem in brain-assisted target speaker extraction.

To address the above issues, this paper proposes a dual-path RWKV-like encoder for EEG signals and integrates it into a brain-assisted target speaker extraction framework. Specifically, dual-path RWKV divides long sequence EEG inputs into spatial paths and temporal paths, corresponding to two complementary modules: a Cortical Interaction Module (CIM) and a Temporal Dynamics Module (TDM). The CIM serves as the spatial path, which models inter-electrode relationships by leveraging learnable spatial shifting operations and decay factors to capture global cortical context; notably, the electrode montage provides a coarse spatial proxy of activities from different cortical regions. The TDM serves as the temporal path, which focuses on modeling the temporal dynamics of EEG signals via efficient gated transformations, enabling robust representation learning under low SNR and non-stationarity. Besides, it is its linear-time complexity with respect to the sequence length, which makes it scalable and practical for long EEG sequences and an increasing number of electrodes.

The experimental results indicate that our framework achieved state-of-the-art performance on three publicly available datasets, provide a new paradigm for efficiently processing long sequence spatiotemporal physiological signals. Our demo is available at[6].

## 2 Related Work

Target speaker extraction (TSE) aims to recover the speech of a desired speaker from a speech mixture by exploiting auxiliary information about the target speaker. Conventional audio-based TSE methods usually rely on reference speech, speaker embeddings, or spatial cues to specify the target speaker. SpeakerBeam extracts a target-speaker representation from an enrollment utterance and uses it to guide speech extraction [28]. VoiceFilter performs speaker-conditioned spectrogram masking with a speaker embedding derived from a reference utterance [29]. Later time-domain systems, such as SpEx and SpEx+, further improve speaker extraction by jointly learning speaker representation and mixture encoding in the waveform domain [30,31]. These methods have shown strong performance and established reference-speech-guided extraction as a representative TSE paradigm.

However, audio-based cues are not always convenient in practical scenarios. Audio-visual target speaker extraction introduces visual information, such as face regions, lip movements, or talking-face dynamics, to identify the target speaker and guide speech extraction. Since lip motion is naturally correlated with speech production, visual cues can provide complementary information when the acoustic mixture is highly overlapped. Early audio-visual speech separation and enhancement studies demonstrate that visual features can help focus the separation network on the desired speaker [32,33]. More recent audio-visual TSE studies and benchmarks further explore how to use synchronized visual cues for target speaker extraction in complex acoustic environments [34,35].

Despite their effectiveness, audio-visual TSE methods still have practical limitations. They require a visible and sufficiently clear face or mouth region of the target speaker, which can be difficult to guarantee in real-world scenarios. EEG-based brain-assisted TSE uses neural responses as auxiliary cues to infer the attended speaker and guide target speech extraction. Neuroscience studies on auditory attention decoding have shown that attended speech can be decoded from neural activity, suggesting that EEG can provide a direct cue about the listener's attentional focus [36]. Early brain-informed methods usually follow a two-stage pipeline, where speech sources are first separated and then matched with EEG-reconstructed speech envelopes to identify the attended stream [37]. BISS integrates reconstructed neural envelopes into a speech separation framework, but it still requires a stimulus reconstruction stage [38]. To simplify this pipeline, BESD directly uses EEG as the reference cue for speech extraction, and U-BESD further improves the encoder–decoder architecture [22,39]. BASEN

[6] https://anonymous2026-is.github.io/TSE/

introduces convolutional multi-layer cross-attention to enhance EEG–speech fusion [21]. MSFNet improves speech and EEG representation learning with multi-scale feature extraction and graph-based EEG modeling [20], while NeuroHeed explores online neuro-steered speaker extraction [40].

Recent brain-assisted TSE studies further emphasize cross-modal alignment and efficient representation learning. M3ANet points out that EEG and speech are not perfectly synchronized because auditory stimuli must pass through the auditory pathway before generating neural responses. It introduces a contrastive modal alignment module to reduce the temporal discrepancy between EEG and speech features, and combines multi-scale convolution with GroupMamba modules for stronger speech representation learning [13].

Although these methods have advanced brain-assisted TSE, robust EEG representation learning remains challenging. EEG signals are low-SNR, high-dimensional, and non-stationary, with attention-related neural responses entangled with physiological artifacts and noise. Existing convolutional, recurrent, or attention-based EEG encoders either have limited ability to capture long-range spatiotemporal dependencies or introduce high computational complexity. In contrast, this work focuses on an efficient EEG representation learner. We propose a dual-path RWKV-like EEG encoder that decomposes EEG modeling into cortical interaction and temporal dynamics paths, enabling the model to capture inter-electrode dependencies and intra-electrode temporal evolution with linear computational complexity.

## 3 Proposed Method

The proposed brain-assisted target speaker extraction framework consists of five modules: an EEG encoder, a speech encoder, a multi-modal loss module, a mask estimator, and a speech decoder, as shown in Fig. 1a. The EEG encoder extracts stimulus-aligned neural features, the speech encoder provides multi-scale audio representations, and the InfoNCE module integrates them to form complementary pairs. The mask estimator then predicts a target speaker mask, and the decoder reconstructs the clean speech waveform.

### 3.1 Dual-path EEG Encoder

The proposed dual-path EEG encoder consists of three components: a Graph Convolutional Network (GCN), residual feature blocks, and a DP-RWKV block built upon the RWKV architecture, as shown in Fig. 1(b). The GCN captures the topological relationships among EEG electrodes, injecting structural priors into feature propagation. The residual blocks enhance hierarchical representation learning and stabilize optimization, alleviating representation degradation in deep encoders.

Due to the scalp electrode montage, EEG signals exhibit strong spatial correlations across channels. However, EEG is also low-SNR, high-dimensional, and

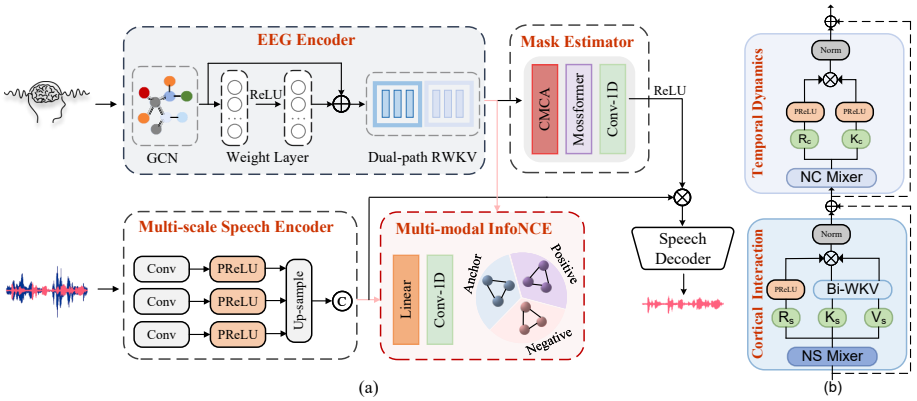



**Fig. 1.** (a) Overall architecture, (b) Dual-path RWKV interaction module.

non-stationary, with attention-related patterns entangled with physiological artifacts and noise. These characteristics hinder efficient long-sequence modeling and joint learning of global inter-electrode interactions and intra-electrode temporal dynamics. To address this, we adapt RWKV [41] for EEG and propose a spatial-temporal dual-path DP-RWKV with linear-time spatiotemporal modeling.

**Spatial Path: Cortical Interaction Module.** The spatial path aims to model global inter-electrode dependencies within each time frame. We treat electrodes as tokens and perform global interaction along the channel axis with linear complexity in $C$. For each time step $t$, we first apply a Neuro-Space Mixer (NS-mixer) along the electrode dimension:

$$\tilde{\mathbf{X}}_s = \mathbf{X}_t + (1-\mu)\cdot \mathbf{X}_t^{\dagger}, \tag{1}$$

where $\mathbf{X}_t^{\dagger}$ is constructed from adjacent electrodes and $\mu$ is a learnable mixing coefficient.

The mixed features are then projected into receptance, key, and value (implemented as linear projections):

$$\mathbf{R}_s = \tilde{\mathbf{X}}_s \mathbf{W}_R^{(s)}, \quad \mathbf{K}_s = \tilde{\mathbf{X}}_s \mathbf{W}_K^{(s)}, \quad \mathbf{V}_s = \tilde{\mathbf{X}}_s \mathbf{W}_V^{(s)}, \tag{2}$$

where $\mathbf{W}_R^{(s)}, \mathbf{W}_K^{(s)}, \mathbf{W}_V^{(s)} \in \mathbb{R}^{C\times C}$ are learnable weights.

To enable efficient global interaction across electrodes, we adopt a bidirectional weighted key–value aggregation (Bi-WKV). We perform two recurrent scans along the electrode token index $c \in \{1,\ldots,C\}$ , preserving linear-time complexity. The Bi-WKV output at electrode token $c$ is defined as:

$$\mathrm{wkv}_c = \frac{a_{c-1} + b_{c+1} + e^{u+k_c}\, v_c}{c_{c-1} + d_{c+1} + e^{u+k_c}}, \tag{3}$$

where $k_c$ and $v_c$ are the $c$-th elements from $\mathbf{K}_s$ and $\mathbf{V}_s$, respectively, and $u$ is a learnable bonus parameter. The forward states $(a_c, c_c)$ and backward states $(b_c, d_c)$ are updated recursively as:

$$\begin{aligned} a_c &= e^{-w/L} \cdot a_{c-1} + e^{k_c} v_c, & c_c &= e^{-w/L} \cdot c_{c-1} + e^{k_c} \\ b_c &= e^{-w/L} \cdot b_{c+1} + e^{k_c} v_c, & d_c &= e^{-w/L} \cdot d_{c+1} + e^{k_c} \end{aligned} \tag{4}$$

where $w$ is a decay parameter and $L$ is a scaling constant.

Finally, the spatial-path output is obtained by receptance gating and a linear projection:

$$\mathbf{O}_{\text{spatial}}(t, :) = \big(\sigma(\mathbf{R}_s) \odot \text{wkv}\big)\mathbf{W}_O^{(s)}, \tag{5}$$

where $\text{wkv} = [\text{wkv}_1, \ldots, \text{wkv}_C] \in \mathbb{R}^{B \times C}$, $\sigma(\cdot)$ denotes the sigmoid function, $\odot$ denotes element-wise multiplication, and $\mathbf{W}_O^{(s)} \in \mathbb{R}^{C \times C}$.

**Temporal Path: Temporal Dynamics Module (TDM).** The temporal path focuses on modeling *intra-electrode temporal dynamics* along the time axis. For each electrode $c$, we take the sequence $\mathbf{X}^{(c)} = \mathbf{X}[:, :, c] \in \mathbb{R}^{B \times T}$ and apply RWKV-style recurrent weighted key–value (WKV) updates along *time tokens* $t \in \{1, \ldots, T\}$, which has linear complexity .

$$\mathbf{O}_{\text{temporal}}(:, t, c) = \big(\sigma(\mathbf{R}_t) \odot \text{wkv}\big)\mathbf{W}_O^{(t)}. \tag{6}$$

**Dual-Path Fusion.** The spatial features $\mathbf{O}_{\text{spatial}}$ and the temporal features $\mathbf{O}_{\text{temporal}}$ are fused with a residual design.

### 3.2 Multi-scale Speech Encoder

The Multi-scale speech encoder employs three parallel 1-D convolutional layers with distinct kernel sizes, designed to extract hierarchical feature representations at multiple temporal scales using a multi-resolution speech representation.

### 3.3 Multi-modal InfoNCE

To align heterogeneous modalities, we adopt a multi-modal symmetric InfoNCE (Information Noise-Contrastive Estimation) loss, which learns a shared embedding space by pulling matched EEG–audio pairs together and pushing unmatched pairs apart within the same temporal window.

For a minibatch containing $K$ paired samples $\{(\boldsymbol{e}_i, \boldsymbol{a}_i)\}_{i=1}^K$, the multi-modal InfoNCE loss for an audio-to-EEG direction is defined as:

$$\mathcal{L}_{a2e} = -\frac{1}{K} \sum_{i=1}^{K} \log \frac{\exp\left(s(\boldsymbol{z}_i^a, \boldsymbol{z}_i^e)/\tau\right)}{\sum_{j=1}^{K} \exp\left(s(\boldsymbol{z}_i^a, \boldsymbol{z}_j^e)/\tau\right)} \tag{7}$$

Symmetrically, the EEG-to-audio loss is defined as:

$$\mathcal{L}_{e2a} = -\frac{1}{K} \sum_{i=1}^{K} \log \frac{\exp\left(s(\boldsymbol{z}_i^e, \boldsymbol{z}_i^a)/\tau\right)}{\sum_{j=1}^{K} \exp\left(s(\boldsymbol{z}_i^e, \boldsymbol{z}_j^a)/\tau\right)} \tag{8}$$

The final contrastive loss is the sum of these two symmetric objectives:

$$\mathcal{L}_{\text{InfoNCE}} = \mathcal{L}_{a2e} + \mathcal{L}_{e2a} \tag{9}$$

### 3.4 Mask Estimator

The cross-modal representation from EEG and speech features is fed into a mask estimation network, implemented with cross-modal cross attention and mossformer blocks [42], an efficient architecture for speech separation.

### 3.5 Speech Decoder

Use the features of the mask generator via a decoder to achieve signal reconstruction by transposed convolution.

### 3.6 Loss Function

The model is optimized using a composite loss function combining reconstruction and contrastive objectives. The reconstruction loss employs SI-SDR loss. And the total loss encompasses InfoNCE loss and SI-SDR loss. The parameter $\lambda$ balances the scales of the different losses. This relationship is encapsulated by the following formula:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{SI-SDR}} + \lambda \mathcal{L}_{\text{InfoNCE}} \tag{10}$$

## 4 Experimental Setup

### 4.1 Datasets

Building on previous studies, we conducted experiments on three large-scale public datasets: Cocktail Party[43] , AVED[44] and MM-AAD dataset[45]. Among them, the Cocktail party dataset uses 128-electrode caps, includes 33 subjects (28 males, 5 females) who complete 30 trials, while others use 32-electrode caps. The AVED dataset includes 20 subjects (14 males, 6 females) who complete 16 trials, and MM-AAD dataset includes 50 subjects (34 males, 16 females) who complete 20 trials.

### 4.2 Data Preprocessing

Raw EEG recordings typically contain substantial noise and artifacts, and thus require preprocessing before modeling. For the Cocktail Party dataset, EEG signals are downsampled to 128 Hz, bandpass-filtered between 0.1–45 Hz, and re-referenced to the mastoid average. Independent component analysis (ICA) is then applied to remove artifacts caused by eye blinks and muscle activities. For the AVED and MM-AAD datasets, we first apply a notch filter to suppress the 50 Hz power-line interference, followed by a 1–50 Hz bandpass filter and

downsampling to 128 Hz. ICA is similarly used to separate mixed components and remove interference, and the EEG signals are finally re-referenced using the common average reference across all electrodes. For all three datasets, the audio stimuli are downsampled to 14.7 kHz. For training and validation, EEG and speech signals are segmented into non-overlapping 2 s clips, while the test set uses 20 s clips.

## 5 Experimental Results

### 5.1 Main results

We evaluated our method against existing brain-assisted speaker extraction approaches on three datasets, achieving superior performance across key metrics as shown in Table 1.

**Table 1.** Performance comparison across different methods on three public datasets.

| Datasets | Methods | SDR (dB) | SI-SDR (dB) | STOI | ESTOI | PESQ |
|---|---|---|---|---|---|---|
| Cocktail Party | Mixture | 0.47 | 0.45 | 74.00% | 55.00% | 1.61 |
| | BSED[22] | - | 5.75 | 79.00% | - | 1.79 |
| | UBESD[39] | - | 8.54 | 83.00% | - | 1.97 |
| | BASEN[21] | - | 12.23 | 86.00% | - | 2.24 |
| | NeuroHeed[40] | -0.09 | -0.11 | 71.48% | 54.79% | 1.45 |
| | MSFNet[20] | 13.03 | 12.89 | 88.00% | 77.00% | 2.51 |
| | M3ANet[13] | <u>14.11</u> | <u>13.95</u> | <u>89.23%</u> | <u>78.36%</u> | <u>2.58</u> |
| | Ours | **15.31** | **15.02** | **90.75%** | **81.45%** | **2.70** |
| AVED | Mixture | 1.54 | 1.52 | 75.83% | 60.57% | 1.54 |
| | UBESD[39] | 8.10 | 7.89 | 85.00% | 72.00% | 1.75 |
| | BASEN[21] | 8.68 | 8.46 | 86.00% | 75.00% | 1.91 |
| | NeuroHeed[40] | 8.77 | 8.61 | 88.11% | 77.81% | 1.82 |
| | MSFNet[20] | 9.84 | 9.65 | 89.00% | 79.00% | 2.07 |
| | M3ANet[13] | <u>11.14</u> | <u>10.89</u> | <u>90.60%</u> | <u>82.06%</u> | <u>2.21</u> |
| | Ours | **12.38** | **12.21** | **91.86%** | **84.40%** | **2.33** |
| MM-AAD | Mixture | 2.14 | 2.12 | 79.67% | 64.10% | 1.54 |
| | UBESD[39] | 6.60 | 6.29 | 84.32% | 69.59% | 1.66 |
| | BASEN[21] | 8.22 | 8.04 | 87.72% | 74.45% | 2.01 |
| | NeuroHeed[40] | 8.42 | 8.03 | 89.10% | 78.10% | 2.18 |
| | MSFNet[20] | 9.69 | 9.57 | 90.19% | 78.71% | 2.18 |
| | M3ANet[13] | <u>10.71</u> | <u>10.57</u> | <u>91.68%</u> | <u>82.31%</u> | <u>2.33</u> |
| | Ours | **11.45** | **11.24** | **92.72%** | **83.61%** | **2.46** |

On the Cocktail Party dataset, our method surpasses M3ANet with gains of 1.20 dB SDR, 1.07 dB SI-SDR, 1.52% STOI, 3.09% ESTOI, and 0.12 PESQ.

On the AVED dataset, our method improves over M3ANet by 1.24 dB SDR, 1.32 dB SI-SDR, 1.26% STOI, 2.34% ESTOI, and 0.12 PESQ, underscoring

the DP-RWKV EEG encoder's effectiveness in enhancing speech quality and reducing distortion.

On the MM-AAD dataset, our method further outperforms prior models with 0.74 dB SDR, 0.67 dB SI-SDR, 1.04% STOI, 1.30% ESTOI, and 0.13 PESQ, demonstrating strong generalizability and efficiency.

### 5.2 Ablation Study

We conducted ablation experiments on the proposed dual path EEG encoder and multi-modal InfoNCE, as shown in Table 2.

**Table 2.** Ablation study results on three public datasets.

| Datasets | Methods | SDR (dB) | SI-SDR (dB) | STOI |
|---|---|---|---|---|
| Cocktail Party | ours | **15.31** | **15.02** | **90.75%** |
| | w/o CIM | 14.36 | 14.11 | 89.56% |
| | w/o TDM | 14.62 | 14.53 | 89.94% |
| | w/o InfoNCE | 15.01 | 14.73 | 89.97% |
| AVED | ours | **12.38** | **12.21** | **91.86%** |
| | w/o CIM | 11.41 | 11.24 | 89.23% |
| | w/o TDM | 11.81 | 11.77 | 89.12% |
| | w/o InfoNCE | 12.11 | 12.01 | 90.87% |
| MM-AAD | ours | **11.45** | **11.24** | **92.72%** |
| | w/o CIM | 10.74 | 10.36 | 91.25% |
| | w/o TDM | 10.92 | 10.88 | 91.65% |
| | w/o InfoNCE | 11.21 | 11.01 | 91.25% |

The experimental results demonstrate that our full model (denoted as ours) achieves the best performance across all evaluation metrics on all three datasets. Specifically, on the Cocktail Party dataset, the full model achieves an SDR of 15.31 dB, an SI-SDR of 15.02 dB, and an STOI of 90.75% . Removing the cortical interaction module (w/o CIM) leads to a significant drop in all metrics, with SDR and SI-SDR decreasing by approximately 1 dB. This underscores the critical role of this module in modeling cross-channel information and spatial dependencies. Similarly, removing the temporal dynamics module (w/o TDM) also causes noticeable performance degradation, particularly a drop of 0.49 dB in SI-SDR, confirming the effectiveness of this module in extracting and refining intra-channel temporal features. Finally, removing the InfoNCE contrastive loss (w/o InfoNCE) results in a slight but consistent decrease across all metrics, indicating that the introduced contrastive learning strategy positively contributes to learning more discriminative representations.

A consistent trend is observed on the AVED and MM-AAD datasets: the full model consistently outperforms any ablated variant. This thoroughly proves that each component in our model — cortical interaction module, temporal dynamics

**Table 3.** Component isolation results on the Cocktail Party dataset.

| ID | EEG encoder | Mask estimator | SI-SDR↑ | STOI↑ |
|---|---|---|---|---|
| S0 | M3ANet[13] | M3ANet[13] | 13.95 | 89.23 |
| S1 | **ours** | M3ANet[13] | 14.68 | 90.10 |
| S2 | M3ANet[13] | **Mossformer**[42] | 14.23 | 89.50 |
| S3 | **ours** | **Mossformer**[42] | **15.02** | **90.75** |

module, and the InfoNCE loss are indispensable. They work collaboratively to contribute to the final performance improvement.

To clearly attribute the independent contributions of each module, we designed an isolation experiment. Table 3 reports a 2×2 component-isolation study to disentangle the contributions of the proposed DP-RWKV EEG encoder and the Mossformer mask estimator. Starting from the baseline system (S0), we replace only the EEG encoder with DP-RWKV while keeping the mask estimator unchanged (S1) to measure the encoder's standalone gain, and replace only the mask estimator with Mossformer while keeping the EEG encoder unchanged (S2) to quantify the effect of the mask estimator. The full system (S3) combines both components. This controlled design ensures that performance improvements can be attributed to the specific replaced module rather than confounding architectural changes.

Concretely, the differences S1–S0 and S2–S0 reflect the isolated gains brought by DP-RWKV and Mossformer, respectively, while S3–S0 captures the overall improvement of combining both. By comparing these deltas, we can verify whether the full-model gains mainly stem from the EEG encoder upgrade, the mask estimator upgrade, or their complementarity. This analysis directly addresses the concern that improvements might be dominated by a stronger separation backbone, and provides a transparent attribution of where the performance gains originate.

## 6 Conclusion

In this paper, we propose a novel dual-path RWKV-like EEG encoder network to better model the low-SNR, high-dimensional, and non-stationary characteristics of EEG signals for target speaker extraction. Our proposed dual-path RWKV encoder effectively captures advanced spatiotemporal-frequency features related to auditory attention through a synergistic dual-path design. The cortical interaction and temporal dynamics modules work in concert to emulate the brain's information processing, integrating global dynamics while enhancing local feature extraction and maintaining linear computational complexity. Future work will focus on improving robustness and transferability, for example via domain adaptation and more data-efficient training strategies.

## 7 Generative AI Use Disclosure

Large Language Models (LLMs) were used solely for manuscript polishing (e.g., rephrasing and grammar checks) to improve clarity and readability. The LLMs were not used for ideation, methodology, experimental design, data analysis, or result interpretation. All scientific content was produced and verified by the authors.

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